BigBurger AI

Enterprise Multi-Tenant Document Intelligence Platform

Upgrading from Regional Silos to Unified Knowledge

Lead Engineer Design Proposal

April 2026 · GCP-Native Architecture

Slide 2 — The Challenge: From Silos to Synthesis

Current State

- 4 independent regional subsidiaries (North, South, East, West)
- Each runs its own Google Drive with PDFs + scanned images
- A single-tenant PoC works for scoped regional questions

Target State

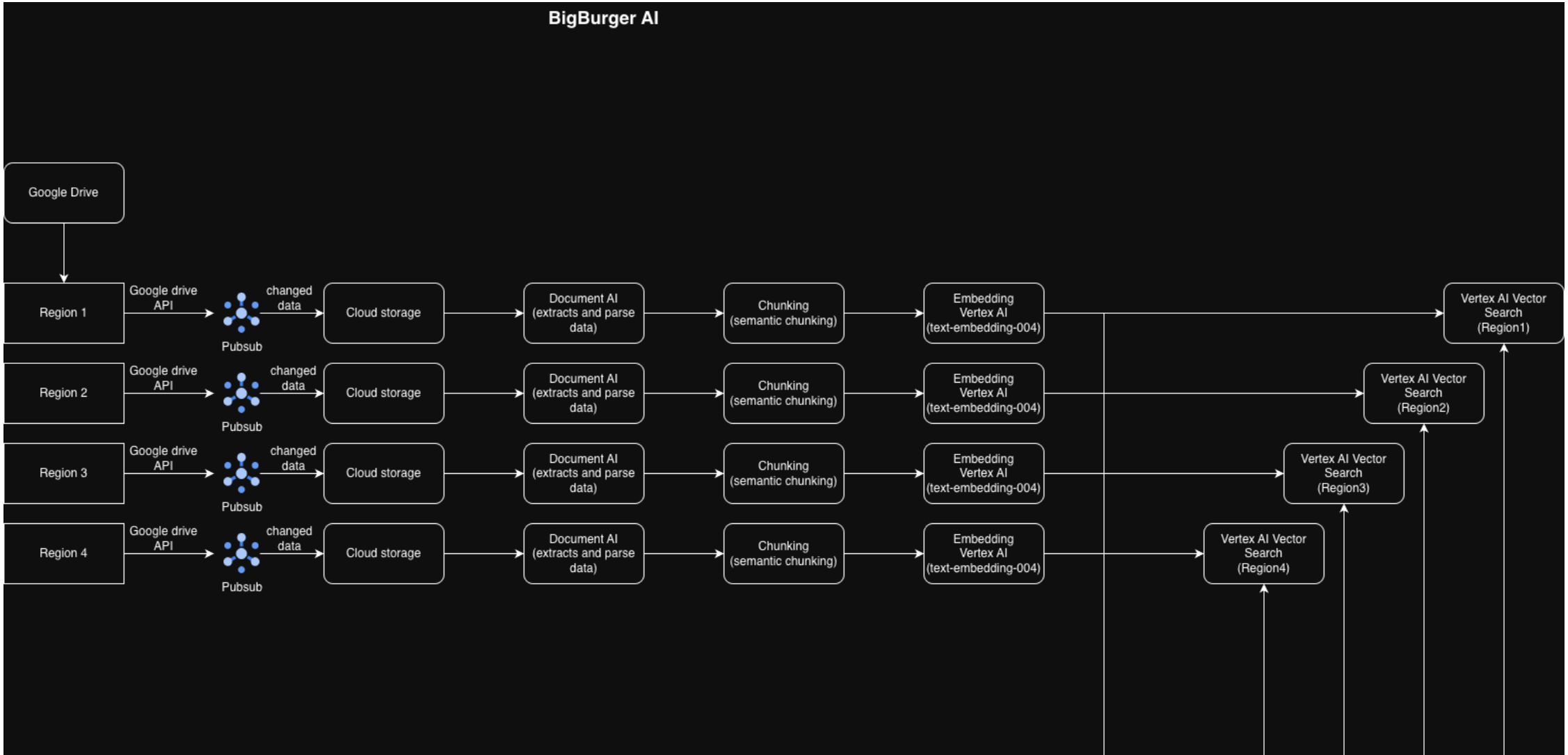
> "Which subsidiaries have conflicting exclusivity clauses with our national beverage supplier, and what is our total aggregated liability?"

Three hard problems this introduces:

1. **Multi-tenancy** — data from 4 siloed Drive environments must be queryable together without data bleed
2. **Cross-regional synthesis** — an LLM must reason across all subsidiaries in one coherent answer
3. **Access control** — regional managers see only their region; global auditors see all

Slide 3 — High-Level System Architecture

High level End to End Architectural Flow



Slide 4 — Pillar 1: Data Ingestion Pipeline

Triggering: Event-Driven Change Detection

- **Google Drive API** webhooks push change notifications to a **Pub/Sub** topic per region
- Cloud Run Job subscribes to the topic and downloads the changed file to **Cloud Storage** (tenant-prefixed bucket: `gs://bigburger-docs/north/`, etc.)
- No polling — near-real-time sync with sub-minute latency

OCR & Extraction

- **Document AI** processes scanned PDFs and images → extracts clean text
- Form Parser extracts structured fields (dates, parties, amounts) for metadata enrichment

Chunking Strategy

- **Semantic chunking** using `RecursiveCharacterTextSplitter` at sentence/paragraph boundaries
  - Chunk size: ~400–600 tokens; overlap: ~80 tokens
  - Preserves legal clause boundaries (critical for accurate retrieval)
- Each chunk carries full provenance metadata: `doc_id`, `region`, `subsidiary`, `doc_type`, `expiry_date`, `liability_amount`, `version`

Embedding

- **Vertex AI** `text-embedding-004` (768 dimensions) — GCP-native, no external API dependency

Slide 5 — Pillar 1 (cont.): Sync, Versioning & Deletions

Document Versioning — Firestore as the Registry

```
Firestore collection: document_registry/{region}/{doc_id}
  {
    doc_id:        "north-bev-001",
    region:        "north",
    drive_file_id: "1aB2cD...",
    current_version: 3,
    ingested_at:   "2026-04-01T10:00:00Z",
    gcs_path:      "gs://bigburger-docs/north/north-bev-001_v3.pdf",
    status:        "active"   // or "deleted"
  }
```

Update Workflow (when a Drive file is modified)

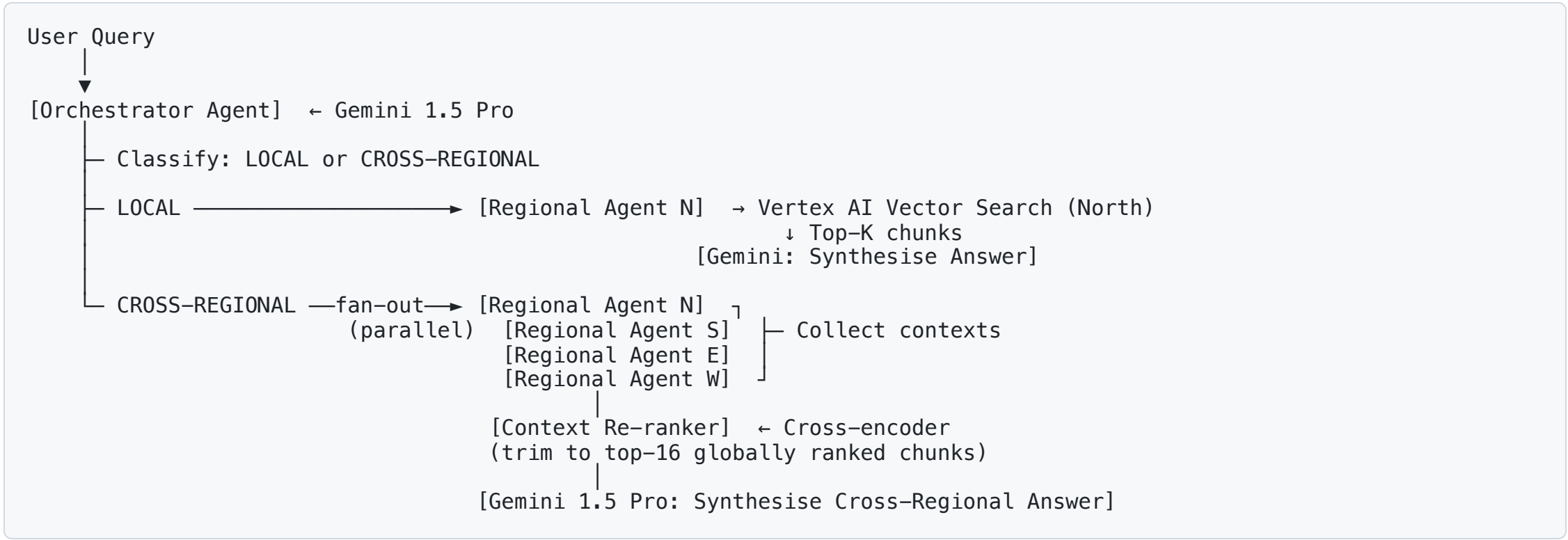
1. Pub/Sub message received → version bump in Firestore
2. New file version downloaded to Cloud Storage
3. Re-chunked and re-embedded → chunks **upserted** by `{doc_id}_{chunk_index}` ID
4. Orphaned chunks from previous version deleted from Vector Search using `doc_id` metadata filter

Deletion Workflow

1. Drive deletion event received → Firestore `status` set to `"deleted"` (soft-delete)
2. Vector Search bulk-delete by `doc_id` metadata filter
3. 30-day tombstone retention for audit trail before hard-delete

Slide 6 — Pillar 2: AI & Agentic Architecture

Two-Tier Agent Design (Google ADK)



Context-Window Exhaustion Prevention

- 4 regions × 4 chunks × ~500 tokens ≈ **8,000 tokens** — far below Gemini's 1M limit
- Cross-encoder re-ranker trims to top-16 chunks when corpus grows
- Structured prompt with explicit region tags prevents hallucination / region confusion

Slide 7 — Pillar 2 (cont.): Token Cost, Rate Limits & Latency

Token Cost Optimization

Query Type	Model	Approx. Cost per Query
Classification (local vs cross-regional)	Gemini 2.0 Flash	~$0.0001
Local retrieval + synthesis	Gemini 2.0 Flash	~$0.001
Cross-regional synthesis	Gemini 1.5 Pro	~$0.005

- Use the **smallest capable model** for each step (Flash for classification, Pro for complex synthesis)
- Cache frequent queries in **Memorystore (Redis)** — 1-hour TTL for identical query+user-role combinations
- Reduce re-embedding cost by caching query embeddings

Rate Limit Handling

- Vertex AI Embedding API: wrap in **Cloud Tasks queue** with rate limiting (requests/min configurable)
- Gemini API: exponential backoff + jitter on 429 responses; queue depth monitored in Cloud Monitoring
- Regional fan-out uses `asyncio.gather` — parallel calls, not sequential, keeping latency ≈ single-region latency

Latency Budget (P95 target: < 4 seconds)

```
Query classification:     ~200ms  (Gemini Flash)
Parallel regional retrieval: ~500ms  (Vector Search ANN)
```

Slide 8 — Pillar 3: Security & Access Control

Authentication (AuthN)

- **Identity Platform** (Firebase Auth) issues signed JWTs
- All API calls require a valid `Authorization: Bearer <token>` header
- Backend (Cloud Run) validates the JWT signature on every request using `firebase_admin.auth.verify_id_token()`
- Tokens expire after 1 hour; refresh tokens managed client-side

Authorization (AuthZ) — RBAC with Custom JWT Claims

```
// JWT custom claims for a regional manager:
{ "role": "regional_manager", "allowed_regions": ["north"] }

// JWT custom claims for a global auditor:
{ "role": "global_auditor", "allowed_regions": ["north","south","east","west"] }
```

Defence-in-Depth (two enforcement layers)

```
Layer 1: Query-time RBAC (Orchestrator Agent)
  → Filter requested regions against JWT allowed_regions before ANY retrieval

Layer 2: Storage-layer isolation (Vertex AI Vector Search)
  → Each region is a separate index — a mis-scoped query physically
    cannot return data from another index
```

Slide 9 — Pillar 4: Observability & Evaluation

Pre-Deployment Accuracy Testing

Golden Dataset (Ground Truth)

- Legal team curates 50–100 QA pairs per region: `{question, expected_answer, source_doc_id}`
- Separate "conflict detection" test set: 20 cross-regional edge cases including known conflicts

RAG Evaluation Metrics (using Vertex AI Evaluation or Ragas)

Metric	Target	Description
Faithfulness	> 0.90	Answer claims are grounded in retrieved context
Answer Relevance	> 0.85	Answer addresses the question asked
Context Precision	> 0.80	Retrieved chunks are actually relevant
Context Recall	> 0.80	Relevant docs are found in the retrieved set
Conflict Detection Rate	> 0.95	Cross-regional conflicts are correctly identified

Regression Suite

- Run full golden dataset after any model update, embedding change, or chunking strategy change
- CI/CD gate: deployment blocked if any metric drops > 5% from baseline

Slide 10 — Pillar 4 (cont.): Production Monitoring & Feedback Loop

Production Monitoring (Cloud Monitoring + Cloud Logging)

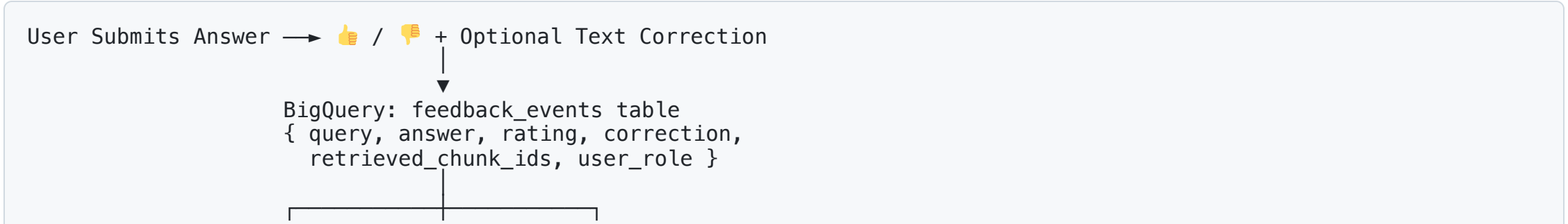
Custom Metrics (emitted from Cloud Run):

- `bigburger/query_latency_ms` — P50, P95, P99 by `query_type`, `user_role`
- `bigburger/retrieval_score` — avg cosine similarity of top-K results
- `bigburger/llm_faithfulness` — sampled automated faithfulness scoring
- `bigburger/auth_rejection_rate` — spikes may indicate breach attempts

Alerting:

- P95 latency > 5s → PagerDuty alert (SLA breach)
- Retrieval score drops > 10% → Slack alert (embedding drift or index corruption)
- Auth rejection rate > 5% in 5 min → Security alert

Continuous Improvement Feedback Loop



Slide 11 — Design Trade-offs

Decision	Choice Made	Alternative Considered	Rationale
Vector DB	Vertex AI Vector Search (one index/region)	Single index with metadata namespace filter	Hard tenant isolation at storage layer; simpler RBAC; slightly higher infra cost
LLM	Gemini 1.5 Pro (synthesis) + Flash (classify)	GPT-4o / Claude	GCP-native = no cross-cloud data egress; tighter IAM integration
Embedding Model	Vertex AI `text-embedding-004`	OpenAI `text-embedding-3-large`	Same GCP-native rationale; 768-dim is sufficient for contract retrieval
Chunking	Semantic / recursive (400–600 tokens, 80 overlap)	Fixed-size (512 tokens)	Legal contracts have clause boundaries; semantic chunking preserves them
Agent Pattern	Supervisor + Sub-Agent (ADK multi-agent)	Single monolithic RAG chain	Fan-out parallelism; each regional agent independently testable; scales to N regions
Data Sync	Pub/Sub webhook + Firestore version ledger	Nightly batch job	Near-real-time update (<1 min vs overnight stale lag)
OCR	Document AI	Tesseract (open-source)	Production-grade accuracy on scanned legal docs; GCP-managed; no self-hosted infra
Auth	Identity Platform + custom JWT claims	Cognito / Auth0	GCP-native; single vendor for identity + audit; integrates with Cloud IAM

Slide 12 — Prototype Overview & What Was Built

Code Prototype: `prototype/`

Demonstrates the hardest part of this problem — **Multi-Tenant Routing & Retrieval** — with:

```
prototype/
├── config.py                    # RBAC roles, mock users (simulates JWT claims)
├── mock_data/contracts.json     # 16 realistic contracts across 4 regions
├── ingestion/
│   ├── chunker.py               # Word-overlap chunker (semantic boundary-aware)
│   └── pipeline.py              # Load → chunk → embed → index orchestration
├── retrieval/
│   ├── vector_store.py          # ChromaDB multi-tenant store (per-region collections)
│   └── retriever.py             # RBAC-enforced retriever
├── agents/
│   ├── rbac.py                  # authenticate() + authorize_region_access()
│   ├── regional_agent.py        # Single-region scoped retrieval agent
│   └── orchestrator_agent.py    # Query classifier + cross-regional fan-out + synthesis
└── main.py                      # 7 live demo scenarios
```

Key Demo Scenarios

1. **Local query** — regional manager, single region, beef contract expiry
2. **Cross-regional conflict detection** — global auditor finds East/BubbleDrink conflict ($1.2M)
3. **RBAC enforcement** — regional manager silently scoped to their region only
4. **Auth rejection** — unknown user ID blocked at authentication layer
5. **Compliance deadlines** — multi-region health inspection schedule query